

The Flow-Free Topological Charge Density: A Task Designed to Separate Gauge-Equivariant Attention from Gauge-Equivariant Convolution

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Abstract

The matched-parameter shootout on the 0^{++} glueball closed on parity: a gauge-equivariant convolutional network (L-CNN) and the gauge-equivariant attention network (GELT) produce the same variational operator to within 1.1σ , at the same mass, with the same signal-to-noise, and the L-CNN does it for a quarter of the cost. This document argues that the outcome was structural rather than accidental, extracts from it a selection rule for tasks on which attention *can* separate, and specifies one such task in full: the prediction of the *gradient-flowed topological charge density* $q_t(x)$ from *unflowed* gauge links. The task is per-site rather than zero-momentum projected, its targets are sparse objects whose position and size vary from configuration to configuration, and its central difficulty — telling a genuine instanton from a lattice dislocation — is a comparison of field strength *across scales at one site*, i.e. a ratio rather than a difference. Softmax attention computes ratios across offsets natively; a gated convolution computes differences and must calibrate them to an amplitude that varies by orders of magnitude across an ensemble. We give the physics, the learning problem, the exact architectural statement of what differs between the two arms, the experimental design with readings fixed in advance, the implementation plan against the existing codebase, and an honest accounting of the ways this can fail. The architecture comparison is a coin flip at roughly 50–55%; the physics deliverable — whether a flow-free learned topological density works at all, and how much of flowed topology is determined by a bounded neighbourhood — does not depend on which arm wins.

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1 Why the glueball shootout tied

1.1 What was measured

The shootout (`notes/lcnn_shootout.md` §9) trained a matched-parameter L-CNN on the identical problem as GELT: same ensemble, same contiguous chain-ordered split, same multi-level smeared input channels, same multi- Δ Rayleigh loss with the same scale pin, same validation-loss checkpoint selection, same blocked jackknife. The match is on four axes — parameters (real degrees of freedom within 10%), receptive field (Manhattan 8 either way), loop degree (≤ 16 after four layers either way), and inputs — and the one asymmetry left standing is the thing under test: L-Conv’s steps are axis-aligned, so the L-CNN reaches off-axis sites only by stacking, whereas GELT’s shortest-path-averaged transport reaches the whole L^1 ball inside one layer.

The result, combining two independently sampled ensembles of 400 held-out configurations each:

	ens0	ens1	combined
$\Delta A_0(\text{GELT} - \text{GEVP})$	$+0.066 \pm 0.031$	$+0.089 \pm 0.030$	$+0.077 \pm 0.022$
$\Delta A_0(\text{L-CNN} - \text{GEVP})$	$+0.062 \pm 0.031$	$+0.072 \pm 0.026$	$+0.068 \pm 0.020$
$\Delta A_0(\text{GELT} - \text{L-CNN})$	$+0.004 \pm 0.008$	$+0.016 \pm 0.013$	$+0.007 \pm 0.007$

with Δm consistent with zero to four decimal places. Both learned operators beat the classical multi-level GEVP by the same margin; neither beats the other.

Parity extends further than A_0 . Measuring the blocked-jackknife noise-to-signal ratio $\delta C(\Delta)/C(\Delta)$ of the two learned operators on the *same* 400 configurations gives

arm	$\Delta = 1$	$\Delta = 2$	$\Delta = 4$	$\Delta = 6$
GELT	0.0322	0.0396	0.0720	0.1695
L-CNN	0.0322	0.0399	0.0728	0.1676

— indistinguishable. This is worth recording because it corrects a natural inference from §9.2 of the shootout note: the observation there that three of nine L-CNN operators concentrate 36–98% of $C(0)$ on a single configuration, where zero of fourteen GELT operators exceed 1.0%, is a statement about *failure rate across training runs*, not about typical variance. It is a reliability argument. It is not an error-bar argument, and no proposal should be built on the claim that attention gives smaller errors on this task.

1.2 The diagnosis

Three properties of the 0^{++} task each independently neutralise attention.

(i) The loss is zero-momentum projected. The Rayleigh quotient $-C(1)/C(0)$ is built from the lattice-summed operator $\hat{O}(t) = \sum_{\vec{x}} O(t, \vec{x})$. Summing over sites is precisely the operation that discards site-to-site variation, which is the one thing an input-dependent kernel has and a fixed kernel does not. `notes/attention_as_operator.md` §1 made this argument to explain why the attention-range statistic ℓ_{att} failed twice — it averaged α over the lattice and therefore measured the learned *kernel*, a design quantity of the same kind as a convolution’s filter width. The identical argument applies to the operator itself. Whatever the attention does differently at site $\vec{x}$ is integrated away before the loss sees it.

(ii) The target is a linear functional of Wilson loops. The operator decomposition (`notes/operator_decomposition.md`; `notes/fable5.1_10-09_audit.md` §8.3) puts only 2.5% of $\|O_{\text{GELT}}\|^2$ outside the span of the 21-operator classical basis, and roughly 80% of that residual is rectangular loop shapes the network rediscovered. The optimum is a fixed linear combination of loop shapes with configuration-independent coefficients. That is exactly what L-Conv is.

(iii) One coupling, one scale, the most symmetric channel. The 0^{++} ground state is the most featureless state in the theory, measured at a single β on a single smearing ladder. A single fixed kernel is optimal by construction; there is nothing to condition on.

Under (i)+(ii)+(iii), softmax attention degenerates into a convolution with learned per-offset weights — which is L-Conv. Parity was guaranteed before the GPU started, and the pre-registration in `reports/curve` §7 said so.

1.3 The selection rule

The architectural difference that survives is narrow and exact. Write a GELT block’s value path and an L-CNN block’s convolution side by side:

$$\text{GEMHSA: } W'(x) = \sum_{\Delta x \in \mathcal{B}_R} \alpha_{\Delta x}(U; x) Q_v^\dagger(x) T_{\Delta x}(x) V(x + \Delta x) T_{\Delta x}^\dagger(x), \quad \alpha = \text{softmax}_{\Delta x} \left(\frac{\text{Re Tr}[Q^\dagger \tilde{K}]}{\sqrt{d_{kv} n_c}} \right), \quad (1)$$

$$\text{L-Conv: } W'(x) = \sum_{\mu, k} w_{\mu k} U_\mu^{(k)}(x) W(x + k\hat{\mu}) U_\mu^{(k)\dagger}(x), \quad w \text{ constant.} \quad (2)$$

It is tempting — and wrong — to summarise this as “L-CNN is a polynomial in the loops, GELT is not”. The L-Act gate

$$W_c \mapsto g\left(\frac{1}{n_c} \text{Re Tr } W_c\right) W_c, \quad g \in \{\text{ReLU}, \text{softplus}\}, \quad (3)$$

is itself *input-dependent*, and because L-Conv mixes channels before it, the gate’s argument is a learned linear combination of transported channels. An L-CNN can therefore realise gates of the form $\text{ReLU}(\text{Re Tr}[A - B])$, i.e. input-dependent *difference* tests between two scales held in different channels. And GELT uses the same L-Act gate on its residual branch. So:

The only architectural difference between the two arms is whether the weights over *offsets* are input-dependent. Gating, bilinearity, receptive field, loop degree, inputs, loss, splits and estimator are matched.

That is a cleaner experiment than the loose version, and it narrows the claim to one mechanism. The mechanism is *normalisation*. The softmax in (1) divides by $\sum_{\Delta x} e^{s_{\Delta x}}$, so α depends only on score *differences* and is invariant under a common additive shift of all scores at a site: it is a relative, scale-free comparison across the neighbourhood, recomputed per site per configuration. The L-Act gate (3) is not: $\text{ReLU}(\text{Re Tr}[A - B])$ fires or not depending on the absolute magnitude of $A - B$, so any decision it implements must be calibrated to an amplitude that the fixed weights cannot adapt.

Hence the selection rule:

*Attention can separate from convolution when the optimal neighbourhood weighting is **relative** rather than absolute — when the right answer depends on the ratio of local structures rather than on their magnitudes, and that ratio varies from site to site and from configuration to configuration.*

Equivalently, a candidate task should have three properties, none of which 0^{++} has: a **per-site or conditional target** (not lattice-summed); **sparse structures at configuration-dependent positions and scales**, so the useful support moves and resizes; and an **amplitude-invariant identity**, so that a fixed-coefficient decision rule must be calibrated and a normalised one need not.

There is already one measurement in this repository consistent with that rule. At $\beta = 0.760$ in 3D $\mathbb{Z}_2$, the coupling closest to β_c in the scan, the classical fixed-radius smeared operator has essentially no signal ($C(1)/C(0) \approx 0.007$) and an untrained network’s attention field has none either, while the *trained* attention field gives a clean exponential (0.496, 0.394, 0.329, 0.276) — see `notes/attention_as_operator.md` §6.1. That is the regime where a fixed kernel stops working and a learned relative one does not. It is suggestive, not decisive: it concerns the attention *field* rather than an output operator, and no L-CNN arm was ever run there.

2 Topology on the lattice

2.1 The continuum object

For a smooth $SU(N)$ gauge field on a compact four-manifold the topological charge

$$Q = \frac{1}{32\pi^2} \int d^4x \epsilon_{\mu\nu\rho\sigma} \text{Tr}[F_{\mu\nu}(x)F_{\rho\sigma}(x)] \in \mathbb{Z} \quad (4)$$

is an integer, and the integrand $q(x)$ — the topological charge density — is a local, gauge-invariant, parity-odd scalar field. Its physics content is substantial: the topological susceptibility $\chi_{\text{top}} = \langle Q^2 \rangle / V$ enters the Witten–Veneziano relation for the η' mass, the θ -dependence of the vacuum energy is controlled by the same distribution, and the two-point function $\langle q(x)q(0) \rangle$ has the celebrated structure of a positive contact term plus a negative-definite tail, which is what makes $\chi_{\text{top}} > 0$ compatible with reflection positivity.

2.2 Lattice discretisations, and one that is in this repository

The Wilson gauge action has no exact topology: the configuration space is connected, and any lattice definition of Q is a real number that approaches an integer only on sufficiently smooth fields. The simplest definition replaces $F_{\mu\nu}$ by the anti-Hermitian part of the plaquette,

$$F_{\mu\nu}^{\text{plaq}}(x) = \frac{P_{\mu\nu}(x) - P_{\mu\nu}^\dagger(x)}{2i}, \quad q_x^{\text{plaq}} = \frac{\epsilon_{\mu\nu\rho\sigma}}{32\pi^2} \text{Tr}[F_{\mu\nu}^{\text{plaq}}(x)F_{\rho\sigma}^{\text{plaq}}(x)], \quad (5)$$

which is exactly `gelt.lattice.topological_charge_density` and exactly Eq. (13) of the L-CNN paper. It carries $O(a^2)$ discretisation errors and a multiplicative renormalisation, $Q^{\text{latt}} \simeq Z(\beta)Q + \text{noise}$ with $Z < 1$ on rough fields.

A defect worth recording. Definition (5) bases all six plaquettes at the *corner* x , so it is not symmetric under reflections about x and is therefore not exactly parity-odd — not even after summing over the lattice. We checked this directly. Build the reflection $x_1 \mapsto (-x_1) \bmod L$ with the correct link relabelling $U'_\nu(y) = U_\nu(Py)$ for $\nu \neq 1$ and $U'_1(y) = U_1(P(y + \hat{1}))^\dagger$. This is an exact symmetry of the Wilson action: on a 6^4 $SU(2)$ configuration the action is preserved to all printed digits, $15478.81704264 \rightarrow 15478.81704264$. Under the same map the charge (5) goes $-1.3161 \rightarrow +2.9888$: the sign flips, the magnitude does not. The remedy is the standard **clover** discretisation, which averages the four plaquettes in each plane around x so that $F_{\mu\nu}$ is centred on the site,

$$F_{\mu\nu}^{\text{clov}}(x) = \frac{1}{8i} \sum_{4 \text{ leaves}} \left(P_{\mu\nu}^{(i)}(x) - P_{\mu\nu}^{(i)\dagger}(x) \right), \quad (6)$$

restoring both the hypercubic symmetry and the exact parity-oddness of q_x and of Q . It does not exist in the repository and is a prerequisite for this study — both for the target and for the sanity checks. It is also the natural home of a new test in `tests/test_lattice.py`: *the clover charge is parity-odd to machine precision and the plaquette charge is not*.

2.3 Dislocations, and why the naive density is not the object of interest

The real problem is not discretisation error but *dislocations*: localised lattice artifacts of size of order a on which $|q_x^{\text{plaq}}|$ is large while nothing topological is present. For the Wilson action the lattice action of an instanton of radius ρ falls below the continuum bound $8\pi^2/g^2$ once $\rho \lesssim a$, so such objects are not Boltzmann-suppressed and contribute a spurious ultraviolet piece to χ_{top} that does not scale. On a thermalised configuration, q^{plaq} is dominated by them.

This has two consequences that matter here. First, the physically meaningful object is not q^{plaq} but the density of a *smoothed* configuration. Second — and this is the whole point of the present proposal — the operation that separates a genuine instanton from a dislocation is intrinsically a comparison *across scales at one site*: is the lump extended over several lattice spacings, or is it a single-plaquette spike? A large $|q_x|$ alone says nothing. The ratio of the field strength averaged at radius a , $2a$, $3a$ says everything. That is a scale-free test, and its threshold cannot be an absolute magnitude, because $|q_x^{\text{plaq}}|$ spans orders of magnitude across an ensemble and across β .

2.4 The gradient flow

The standard smoother is the Wilson (gradient) flow,

$$\dot{V}_t(x, \mu) = -g_0^2 \partial_{x,\mu} S_W[V_t] V_t(x, \mu), \quad V_0 = U, \quad (7)$$

a deterministic diffusion whose effect at flow time t is to average the field over a physical radius

$$r_{\text{sm}} = \sqrt{8t}. \quad (8)$$

On the lattice the drift is read straight off the staple. With $A_\mu(x)$ the staple sum in the repository's convention — so that the local action is $S_{\text{local}} = -(\beta/n_c) \text{Re Tr}[U_\mu(x)A_\mu(x)]$ — one integrates

$$U_\mu(x) \longleftarrow \exp(\varepsilon Z_\mu(x)) U_\mu(x), \quad Z_\mu(x) = -\left[U_\mu(x) A_\mu(x) \right]_{\text{TA}}, \quad (9)$$

where $[M]_{\text{TA}} = \frac{1}{2}(M - M^\dagger) - \frac{1}{2N} \text{Tr}(M - M^\dagger)\mathbb{I}$ is the traceless anti-Hermitian projection. For $n_c = 2$ both $[\cdot]_{\text{TA}}$ and the exponential are closed-form — $\exp(i\theta \hat{n} \cdot \vec{\sigma}) = \cos \theta + i \sin \theta \hat{n} \cdot \vec{\sigma}$ — so no `linalg` kernel is needed, in keeping with the repository’s existing $SU(2)$ fast paths.

The flow is *not* a linear smoothing. In the small-field regime (7) linearises to a heat kernel of radius $\sqrt{8t}$, but on thermalised fields at $\beta \sim 2.4$, where the mean plaquette is far from unity, it is strongly nonlinear, and the nonlinearity is exactly what removes dislocations: a sub-lattice-size instanton shrinks under the flow and falls through the lattice, while a genuine one of radius $\rho \gg a$ is preserved. That asymmetry — destroy the spike, keep the lump — is the part of the map that a linear filter cannot represent, and it is the part on which this experiment rests.

2.5 Why the *density* and not just Q

Predicting the single integer Q_t is a substantially easier and largely solved problem (§5). The field $q_t(x)$ is a strictly richer object and it is what the physics wants: the instanton size distribution, which at one loop behaves as $n(\rho) d\rho \propto \rho^{b-5} d\rho$ with $b = 11N/3$ (so $\rho^{7/3}$ for $SU(2)$) until infrared effects cut it off; the local structure of the vacuum (instanton-liquid versus sign-coherent sheets); and the two-point function $\langle q(x)q(0) \rangle$, whose contact term and negative tail are the subject of a long literature and which is badly distorted by the very smoothing used to define it. A learned map from $\tau = 0$ gives, in principle, a density at the original lattice spacing — something flow cannot provide by construction.

3 The learning problem

3.1 Definition

Fix a gauge group $SU(2)$, a coupling β , a lattice $\Lambda = L^4$, and a flow time t . Let $\mathcal{F}_t$ denote the Wilson flow (7) integrated to t . The task is to learn

$$\hat{q} : U \mapsto \hat{q}(x)[U] \approx q_t(x) \equiv q_x^{\text{clov}}[\mathcal{F}_t(U)], \quad U \text{ a thermalised configuration at } \tau = 0, \quad (10)$$

by minimising a per-site mean-squared error over the lattice and the ensemble. The prediction of the total charge follows as $\hat{Q} = \sum_x \hat{q}(x)$.

Three things about (10) deserve emphasis. The input is **unflowed** — the raw Monte Carlo configuration. The target is the **density field**, not a scalar. And the map is a genuine inference: the network must perform, implicitly and in one shot, the work that the flow does over hundreds of integrator steps.

3.2 Is it well posed?

Topological charge is a global invariant, so it is worth being explicit about why a *local* network can learn $q_t(x)$ at all. The flow is a parabolic equation; its Green function is exponentially localised on the scale $r_{\text{sm}} = \sqrt{8t}$. Consequently $q_t(x)$ depends on the unflowed links in a neighbourhood of x of radius $O(\sqrt{8t})$ up to exponentially small tails. The task is therefore well posed as a local regression *provided*

$$\sqrt{8t}/a \lesssim R_{\text{net}}, \quad (11)$$

where R_{net} is the network’s receptive field — Manhattan 8 for both arms at $R = 2$, $K = 2$ and four layers. Condition (11) is not a nuisance; it is a *measurement*. Sweeping t and watching where each architecture’s accuracy collapses answers the physics question “how much of flowed topology is determined by a bounded neighbourhood of the unflowed field?”, and doubles as a capacity control (§6.7).

Note also that no network can be expected to reproduce the integer quantisation of Q_t exactly, since that is a global property emerging from a sum of local predictions. How close $\hat{Q}$ comes to an integer is a diagnostic of quality, not a constraint that is imposed.

3.3 What this is not

It is important to distinguish (10) from the topology task in the L-CNN literature, because the names collide. Favoni et al. regress q_x^{plaq} of the configuration the network is shown. That is a local algebraic identity: (5) is a degree-two polynomial in plaquettes at a single site, and a single L-Bilin layer spans it exactly. Their demonstration applies the trained model *along* a Wilson flow trajectory — at each flow time it is fed the flowed links and asked for the flowed naive density — and the integer values in their figure come from the flow, not from the network. The map is input-to-same-configuration. Ours is input-at- $\tau = 0$ -to-target-at- $\tau = t$: nonlocal, nonlinear, and scale-selective. The L-CNN’s proven competence at the former says very little about the latter, which is precisely why the comparison is worth running.

4 Why this task can discriminate

Against the selection rule of §1.3, the task scores on all three properties.

Per-site target. The loss is $\sum_x (\hat{q}(x) - q_t(x))^2$. Nothing is summed over the lattice before the gradient is taken, so the site-to-site variation that the 0^{++} Rayleigh quotient integrated away is exactly what is being supervised. Tie-cause (i) is removed by construction.

Sparse structures at configuration-dependent positions and scales. Instantons are dilute, localised, and distributed in size. Within a single ensemble ρ/a already spans a range, and across the β ladder the mean ρ/a moves by roughly a factor of two. The neighbourhood that the network must aggregate over is therefore not fixed: it is $\rho(x)$, a quantity that varies within a configuration and across an ensemble. Tie-cause (iii) is removed.

Amplitude-invariant identity. This is the load-bearing one. The decision “is this lump topological or is it a dislocation?” is invariant under rescaling the local field strength — it depends on the *shape* of the profile, i.e. on ratios of field strength at radius a , $2a$, $3a$, not on their absolute size. Now compare the two architectures’ available decision rules at one site:

$$\text{GELT: } \alpha_{\Delta x} = \frac{e^{s\Delta x}}{\sum_{\Delta x'} e^{s\Delta x'}} \quad \text{— invariant under } s \rightarrow s + c, \text{ a pure ratio across offsets;} \quad (12)$$

$$\text{L-CNN: } g(\text{Re Tr}[\sum_c w_c W_c]) \quad \text{— a threshold on an absolute magnitude.} \quad (13)$$

An L-CNN can hold a radius- a average in one channel and a radius- $3a$ average in another, and it can gate on their difference. What it cannot do with fixed weights is gate on their *ratio*: the threshold in $\text{ReLU}(\text{Re Tr}[A - B])$ is an absolute number, so the decision is correct only for configurations whose local field strength sits near the scale the weights were calibrated to. Since $|q_x^{\text{plaq}}|$ is heavy-tailed within an ensemble and shifts systematically with β , that calibration cannot be right everywhere.

This is a statement about efficiency, not about representability. With enough channels and depth an L-CNN can approximate a ratio on any bounded domain; the claim is that at a matched budget of ~ 15 – 26 k real degrees of freedom it will spend capacity on a filter bank plus a calibrated selector where GELT gets the selector for free from the softmax. That is a bet, and §8 prices it honestly.

5 Prior art, and what is open

work	input $\rightarrow$ target	status
Favoni, Ipp, Müller, Schuh, <i>PRL</i> 128 (2022) 032003	flowed links $\rightarrow q^{\text{plaq}}$ of <i>the same</i> configuration	solved; local, degree-two, algebraic. One L-Bilin layer spans it.
Favoni, Ipp, Müller (2022), arXiv:2212.00832	links $\rightarrow$ links, neural-ODE Wilson flow	toy models only; a different problem (learning the flow map itself).
Matsumoto et al., <i>PTEP</i> 2021 023D01 ($SU(3)$)	q -density at <i>small</i> flow time $t/a^2 \leq 0.3 \rightarrow$ integer Q at large t	$> 99\%$; scalar target, partial flow, non-equivariant networks. Best results integrate out space.
this proposal	unflowed links ($\tau = 0$) $\rightarrow$ flowed density field $q_t(x)$	open.

Two honest consequences follow. First, the gap is real: nobody has learned the flowed *density field* from a genuinely unflowed configuration, and the L-CNN’s published topology success is on the algebraic task, not this one. Second, the *compute-saving* motivation is weaker than it first appears. Matsumoto et al. show that a little flow plus a simple classifier already recovers the integer Q with high accuracy, so “skip the expensive flow” is partly taken. The motivation that survives is the density field itself (§2.5) and the locality question (11). The write-up must lead with those, not with a cost argument it cannot fully support.

6 Experimental design

6.1 Ensembles and scale

Three couplings, with the lattice size chosen so that the physical volume is approximately constant, exploiting the fact that both architectures are translation-equivariant and can be trained and evaluated at any size:

β	$a\sqrt{\sigma}$ (lit.)	a [fm]	L	La [fm]	configurations
2.3	≈ 0.369	≈ 0.165	8	1.32	1200
2.4	≈ 0.266	≈ 0.119	12	1.43	1200
2.5	≈ 0.188	≈ 0.084	16	1.35	1200

The $a\sqrt{\sigma}$ values are the standard Wilson-action $SU(2)$ numbers and are quoted here only to fix the design; **they must be re-derived in-repo** from `rectangular_wilson_loop` before any physical statement is made, in keeping with the repository’s practice of never importing a scale it has not measured. The lattice spacing changes by a factor of two across the ladder, so ρ/a — the object size in the units the network sees — changes by the same factor. That is the scale-adaptivity axis.

Sampling uses the existing exact $SU(2)$ heat-bath plus overrelaxation (`heatbath_overrelaxation_sweep`), which is the prerequisite for decorrelated topology: topological modes have the longest autocorrelation time in the theory, and n_{skip} must be set from `integrated_autocorrelation_time` applied to Q itself, not to the plaquette. **Topological freezing is a live risk at $\beta = 2.5$** and must be measured, not assumed: if $\tau_{\text{int}}(Q)$ is large the effective statistics collapse and the $\beta = 2.5$ row becomes a single-sector dataset, which would invalidate the aggregate readings while leaving the per-site ones intact.

6.2 Flow, and the target ladder

The flow (9) is integrated at fixed **lattice-unit** flow time t/a^2 , identical across β . This is deliberate and it is the design decision that makes the experiment interpretable. Fixing t/a^2 holds r_{sm}/a — and therefore the receptive-field requirement (11) — constant across the ladder, so that any degradation of a mixed- β model is attributable to *scale adaptivity* and not to a capacity wall that bites at one β and not another. Fixing t physically would confound the two.

The primary target is $t/a^2 = 2$, i.e. $r_{\text{sm}}/a = 4$, comfortably inside Manhattan 8. The full ladder

$$t/a^2 \in \{0.5, 1, 2, 4, 8\}, \quad r_{\text{sm}}/a = \{2.0, 2.8, 4.0, 5.7, 8.0\} \quad (14)$$

is generated once per configuration and stored, because it costs one flow trajectory and yields five targets. The largest rung deliberately sits at the receptive-field boundary.

Gate 0. Before any training, verify the $Q(t)$ plateau at every β : flow to $t/a^2 = 16$, plot $Q(t)$, and confirm that $t/a^2 = 2$ lies on a plateau and that Q is within a stated tolerance of an integer. If it does not at $\beta = 2.3$ — which is plausible, the coarsest lattice being the one where dislocations survive longest — the primary rung moves, and the fact is recorded rather than hidden.

6.3 Splits

Contiguous and chain-ordered, matching the glueball protocol: 800 train, 200 validation, 200 test, with the test block at the far end of the Markov chain. This is deliberately the harder split; `notes/lcnn_shootout.md` §9.1 shows why it matters (a network can fit chain-local structure that survives into an adjacent validation block and not into a distant test block). Per-site supervision gives $800 \times L^4$ training targets, so the statistics per configuration are enormous and small batches suffice.

6.4 Matched budgets

The four-axis match of `notes/lcnn_shootout.md` §2 carries over, with $D = 4$ substituted:

axis	GELT	L-CNN
receptive field	$R = 2 L^1$ ball $\times$ 4 layers $\rightarrow$ Manhattan 8	$K = 2$ symmetric $\times$ 4 layers $\rightarrow$ Manhattan 8
loop degree	bilinear value path, ≤ 16	L-CB doubling, ≤ 16
inputs	6 plaquette channels ($n_{\text{pairs}} = D(D-1)/2$)	identical
parameters	tune d_{model}	tune c_{hidden} to within 15%
output	<code>reduction="none"</code> $\rightarrow (B, \Lambda)$	<code>reduction="none"</code> $\rightarrow (B, \Lambda)$

Two $D = 4$ specifics. The L^1 ball at $R = 2$ in four dimensions has 40 offsets against the glueball task’s 24 in three, so the transport stage — already 62.8% of a GELT step (`notes/performance_audit.md` §5.0) — grows by $1.67\times$ per site. And **RoPE requires** $d_{qkv} \geq 2D = 8$ (known caveat 2): below that, whole lattice axes sit on the identity rotation and the score stops being offset-selective along them. This is a hard constraint on the GELT arm’s geometry and must be asserted in the constructor, not remembered.

6.5 Arms

Six arms, of which two are classical and cost nothing:

1. **Identity:** $\hat{q} = q_x^{\text{clov}}[U]$, the unflowed density. The zero-work baseline; everything must beat it.
2. **Best linear filter:** the least-squares-optimal isotropic linear filter of support Manhattan 8 applied to $q^{\text{clov}}[U]$, fitted on the training split. This is the “a convolution is the right answer” hypothesis made concrete and is the single most important control in the design: if the networks do not clearly beat it, nothing nonlinear has been learned and the architecture comparison is moot.
3. **GELT, per- β** (three models).
4. **L-CNN, per- β** (three models).
5. **GELT, mixed- β** (one model trained on all three ensembles).
6. **L-CNN, mixed- β** (one model).

Plus untrained controls of both architectures at three seeds each, following the precedent of `notes/fable5.1_10-09_audit.md` §8.3, with the seed-aggregation rule (median, not mean) fixed *before* the numbers are seen — §9.2 of the shootout note records why.

6.6 Metrics

- M1 — per-site accuracy.** $1 - R^2$ on the test split, i.e. $\sum_x (\hat{q} - q_t)^2 / \sum_x (q_t - \langle q_t \rangle)^2$. Blocked jackknife over configurations, block size from $\tau_{\text{int}}(Q)$.
- M2 — charge accuracy.** $\text{RMS}(\hat{Q} - Q_t)$ and the RMS distance of $\hat{Q}$ to the nearest integer.
- M3 — physics.** χ_{top} and the histogram of Q reconstructed from $\hat{q}$ versus from q_t , on the same configurations.
- M4 — stratified error.** M1 restricted to the dislocation-rich tail: sites in the top percentile of $|q^{\text{clov}}[U]|$, and configurations in the top decile of roughness. *This is where the mechanism predicts the separation, and it is the reading to look at first.*
- M5 — the two-point function.** $\langle \hat{q}(x)\hat{q}(0) \rangle$ against $\langle q_t(x)q_t(0) \rangle$, including the sign of the tail.
- M6 — robustness.** Top-1 configuration share of the loss, under the uniform gate installed in `fit_glueball_overlap.py` by `notes/lcnn_shootout.md` §9.2.

All GELT-versus-L-CNN differences are taken *inside* each jackknife sample on shared configurations, as `fit_glueball_overlap.py` -vs does: the two arms are correlated through the ensemble, and the correlated error measured roughly $4\times$ smaller than the naive one on the glueball task.

6.7 Readings, fixed in advance

- R1 (primary, architecture).** $\Delta(1 - R^2)$ between mixed- β GELT and mixed- β L-CNN, correlated jackknife. **GELT wins** if the relative error is lower by more than 2σ . **Parity** if within 1σ .
- R2 (control, adaptivity).** The same difference at fixed β . *Prediction: parity, or a margin clearly smaller than R1's.* If single- β already separates, the effect is not scale adaptivity and the claim must be re-stated — possibly in GELT's favour, but for a different reason.

- R3 (control, capacity).** Accuracy across the flow-time ladder. Both arms must collapse as $r_{\text{sm}}/a \rightarrow 8$. If GELT's collapse sets in later, that is receptive-field efficiency, not adaptivity, and must be reported as such.
- R4 (physics, architecture-independent).** Does *either* network reproduce χ_{top} and the Q -histogram from $\tau = 0$ within the statistical error of the flowed reference? This is the deliverable that does not depend on who wins.
- R5 (nulls).** The untrained arms must fail. The identity arm and the best-linear-filter arm must be clearly beaten. If arm 2 is not beaten, stop: the task is a convolution and the paper says so.

None of the five outcomes is a wasted run, which is the argument for spending the time — the same argument §7 of the shootout note made, and it held.

7 Implementation

7.1 What already exists

More than one would expect. Verified by direct execution at $D = 4$, $SU(2)$, $L = 6$, `complex128`:

- `plaquette_tensor` returns $(B, 6, \Lambda, 2, 2)$ in $D = 4$; `topological_charge_density` returns (B, Λ) ; `build_transport_average` returns $(B, 40, \Lambda, 2, 2)$ at $R = 2$; `build_axis_transports` returns $(B, 4, 2, \Lambda, 2, 2)$.
- Both models run and are gauge invariant to machine precision: drift 0.0 (GELT), 1.4×10^{-14} (L-CNN), against 1.5×10^{-16} for the target itself.
- **LCNN already accepts `reduction="none"`** and returns (B, Λ) , so no per-site head has to be written for the baseline.
- `staple_sum(..., batched=True)` supplies the flow drift (9) directly, already vectorised over the configuration axis.
- `data.build_plaquette_datasets` takes `target` as a callable `target(configs, group) -> Tensor`, so the flowed density binds in with `functools.partial` with no change to the builder, and R makes it emit (X, T, y) triples with the transport precomputed.
- `integrated_autocorrelation_time` sets n_{skip} .

7.2 What must be written

1. `gelt/flow.py` — `wilson_flow(U, group, t, eps, integrator)` implementing (9), with the closed-form $SU(2)$ exponential and the traceless anti-Hermitian projection, batched over configurations, plus `flow_trajectory` returning the target ladder in one pass. Lüscher's third-order Runge–Kutta is preferred over Euler; Euler at small ε is the fallback and the cross-check.
2. `topological_charge_density(..., definition="clover")` in `gelt/lattice.py`, per (6). **Required**, not optional: §2.2 shows the existing plaquette definition is not parity-odd.
3. `scripts/train_topology.py` — the trainer, modelled on `train_glueball.py`'s `single_build_model()` discipline so that `TOPO_ARCH=lcn` switches the architecture and nothing else, and so that a profiler cannot drift from the training path. Per-site MSE, validation-loss checkpoint selection, artifact names carrying β , L , flow time, architecture, width and seed.

4. `scripts/measure_topology.py` — the classical arms (identity, best linear filter), gate 0's $Q(t)$ plateau, $\tau_{\text{int}}(Q)$, and the ensemble cache.
5. `scripts/fit_topology.py` — offline: M1–M6 from the dumps, with the correlated `-vs` difference.
6. Tests: clover parity-oddness to machine precision (and the plaquette definition failing it); flow gauge covariance; flow preserving the group; Euler-versus-RK3 agreement; the $D = 4$ parameter match between the two arms; the $d_{qv} \geq 2D$ assertion.

7.3 Cost

Estimated from the measured V100 numbers in `notes/performance_audit.md` (GELT 5.04 s/step at 248 832 sites $\times$ 24 offsets; the matched L-CNN 1.29 s/step):

phase	scale	estimate
ensembles (3β , heat-bath + OR)	8^4 , 12^4 , 16^4 , 1200 each	~ 1.5 nights
flow targets (5-rung ladder)	3600 trajectories	1–2 hours
GELT trainings	3 per- β + 1 mixed	~ 2 nights
L-CNN trainings	3 per- β + 1 mixed	~ 0.5 night
untrained + classical arms	offline	hours
total		~ 4 – 5 V100 nights, ~ 1 build week

8 Risks, and what each outcome means

The map is mostly a smoothing. The strongest objection. In the small-field limit the flow *is* a linear filter, so a convolution is the right architecture for the bulk of the map and the adaptive part is a correction. The size of that correction is an empirical question, and arm 2 (best linear filter) measures it directly: the gap between arm 2 and the best network *is* the nonlinear content of the flow. If that gap is small, the whole study reduces to “the flow is nearly linear at these couplings”, which is itself a clean, quotable result — and it terminates the architecture question honestly rather than burying it.

The L-CNN approximates the ratio well enough. At four layers and five or six channels with L-Act gates it may build a serviceable calibrated selector. *Probability I assign to a clear GELT win on R1:* 50–55%. This is down from the 65% quoted before this audit, for three reasons: the surviving mechanism is one axis rather than four (§1.3); the map is mostly a smoothing; and the base rate in this repository for “attention will help here” predictions is currently zero for one.

Topological freezing. At $\beta = 2.5$ the charge may decorrelate so slowly that the 1200 configurations contain a handful of independent topological samples. This does not affect the per-site readings M1/M4, which have L^4 targets per configuration, but it would invalidate M2/M3 at that coupling. Mitigation: measure $\tau_{\text{int}}(Q)$ first and report it; if freezing bites, quote the aggregate physics at $\beta = 2.3$ and 2.4 only, and say why.

Single- β already separates. The instanton size distribution is broad within one ensemble, so the adaptivity requirement may not need the β ladder at all. This breaks R2 as a control while strengthening the overall case. The response is to re-stratify by measured ρ rather than by β and to say plainly that the pre-registered control did not behave as predicted.

The target is ill-defined at the coarsest coupling. If $Q(t)$ has no plateau at $\beta = 2.3$ and $t/a^2 = 2$, the regression target is partly arbitrary. Gate 0 catches this before any GPU time is spent.

GELT's cost. The transport is already 62.8% of a GELT step in three dimensions and grows by $1.67\times$ in four. If the step is intolerable, `notes/performance_audit.md` §5.1 — the adjoint $SO(3)$ representation for $SU(2)$, worth a predicted $\sim 3.5\times$ on the transport and $1.8\times$ end to end — is the lever, and this task is a better reason to pull it than the glueball one was.

What each outcome buys. A GELT win on R1 with parity on R2 is the designed result: attention buys scale adaptivity, and the thesis has the architectural claim the 0^{++} chapter could not supply. Parity on both is a second null, and after the shootout a second null is genuinely informative — it would say that input-dependent offset weighting does not pay on *local regression* either, which sharply narrows where the mechanism could ever pay and is worth writing down. An L-CNN win points at the transport average or the receptive field rather than at the loss, and should be reported before an examiner finds it. In all three cases R4 stands on its own.

9 Summary

The 0^{++} shootout tied because the task had none of the three properties that let an input-dependent kernel pay: its loss was zero-momentum projected, its optimum was a linear functional of loop shapes, and it lived at a single scale. Prediction of the gradient-flowed topological charge density from unflowed links has all three. It is per-site; its targets are sparse objects whose position and size vary within and across ensembles; and its central operation — separating a genuine instanton from a lattice dislocation — is a scale-free ratio test, which is what a softmax over offsets computes natively and what a gated convolution must calibrate. The architectural difference between the two arms reduces, exactly, to whether the offset weights depend on the input, which makes the experiment a controlled A/B on one mechanism.

The honest odds on the architecture question are close to even. The reason to run it anyway is that the physics deliverable is separable and largely independent of the outcome: no one has learned the flowed topological charge *density* from a genuinely unflowed configuration, and the flow-time ladder turns the receptive-field constraint into a measurement of how much of flowed topology is determined by a bounded neighbourhood of the raw field. Both are worth having whichever architecture wins.